Finding the Nuclei That Models Miss

Reliable instance segmentation and classification
across diverse tissues in PanNuke

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Project 5

Zhiyu Song, Yuchen Ai, Xinhao Jin, Zheng Jian

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Project summary

Missed nuclei bias measurements of tissue composition even when average segmentation scores are high. In our PanNuke experiments, the largest detection deficit occurs in the Dead class, which includes apoptotic and necrotic nuclei. Small nuclei and instances truncated by image boundaries warrant particular attention.

The proposed study examines both training supervision and instance representation. A controlled experiment tests whether retaining geometric targets for cropped nuclei improves boundary detection; PM-Net replaces horizontal/vertical distance maps and watershed decoding with point-conditioned queries that predict individual masks and cell types. Matched comparisons will assess detection gains against segmentation quality and a twofold inference-latency limit.

Intellectual merit

By varying training targets and decoder design separately under a shared encoder and evaluation protocol, this study will assess their contributions to rare-nucleus detection and mask quality. The resulting ablations will connect changes in aggregate scores to errors in locating, separating, and typing individual nuclei.

Broader impacts

Reliable nucleus counts and cell labels support quantitative studies of tissue composition and the tumor microenvironment. Shared evaluation code and tissue-specific error analyses will make improvements in rare-class detection easier to assess alongside segmentation quality and computational cost.

1. Research aims, context, and preliminary evidence

Nuclear appearance in hematoxylin and eosin (H&E) images varies with staining, morphology, and tissue context, while densely packed nuclei are difficult to separate. Misses, merges, and type confusion can bias cell-composition estimates, particularly for rare apoptotic and necrotic nuclei. PanNuke provides instance masks and five cell classes across 19 tissues for studying these errors [1].

Specific aims

Aim 1 - Characterize segmentation and typing failures. Consolidate HoVer-Net and CellViT-UNI results under a shared evaluator, separating missed nuclei, merges, splits, and type errors by tissue and class.

Aim 2 - Evaluate supervision for truncated nuclei. Complete the four-arm, two-seed comparison of stronger cropping and selective HV supervision, using both fixed crop windows and original validation images.

Aim 3 - Develop a point-conditioned instance model. Test whether PM-Net improves Dead detection and crowded-instance separation within predefined quality and inference-cost limits.

Prior work and architectural motivation

HoVer-Net separates instances using horizontal/vertical (HV) offset maps [2], while CellViT combines a transformer encoder with nuclei segmentation and classification [3]. Our reproduced CellViT-UNI is the primary baseline. PM-Net draws on point-conditioned masks [4], query-based set prediction [5], and spatial adaptation [6] to predict nuclei directly. Matching the pretrained UNI encoder [10] and unfreezing schedule, with parameter-matched adapter controls, will help distinguish decoder effects from backbone and capacity differences.

Data and baseline evidence

PanNuke contains 7,901 RGB patches at 256 x 256 pixels and roughly 200,000 annotated nuclei. Its three prepared folds contain 2,656, 2,523, and 2,722 patches. Each instance is labeled neoplastic, inflammatory, connective, dead, or epithelial. Development uses fold 1 for training and fold 2 for validation.

Reproduced baselines: three-split means	mPQ	bPQ
HoVer-Net, final checkpoint, no test-time augmentation	0.4564	0.6613
CellViT-UNI, final checkpoint, no test-time augmentation	0.4995	0.6653

Earlier analysis found that 38.5% of labeled Dead nuclei had a maximum intersection-over-union (IoU) below 0.1 with model predictions, indicating a substantial detection deficit. Reclassification, augmentation, and frozen-feature probes yielded no consistent improvement [7]. Combining two prediction scales modestly improved Dead PQ at about 6.1 times the inference cost, motivating a more efficient approach to instance detection.

2. Aim 2: supervision for truncated nuclei

Spatial cropping truncates nuclei at training-image boundaries. For instances smaller than 30 pixels, the existing target generator sets HV values to zero. Aim 2 tests whether retaining these geometric targets for newly truncated fragments improves detection; matched intervention arms use identical images and other labels.

Arm	Crop probability	HV target rule
A0: retrained baseline	0.1	Minimum area 30
A1: crop-only control	0.5	Minimum area 30
A2: selective supervision	0.5	Minimum 1 for newly truncated instances with visible area below 30
A3: small-object control	0.5	Minimum 1 for all instances; targets differ only below area 30

All arms use seeds 19 and 1, 130 epochs, matched initialization and optimization, and final-checkpoint evaluation at native resolution without test-time augmentation (TTA). A1-A3 share the augmentation random sequence. Instance identities before and after cropping distinguish newly truncated objects from those already cut by a patch boundary; areas are measured after geometric transformations [8].

Interim findings

An interim development snapshot includes six completed fold-2 evaluations: all four arms for seed 19, and A0/A1 for seed 1. Relative to A0, A2 seed 19 improves Dead PQ by 0.0222 and strict Dead PQ by 0.0025, but reduces mPQ by 0.0091 and bPQ by 0.0062. These losses exceed the predefined 0.002 tolerance, so this configuration does not qualify for expansion [8].

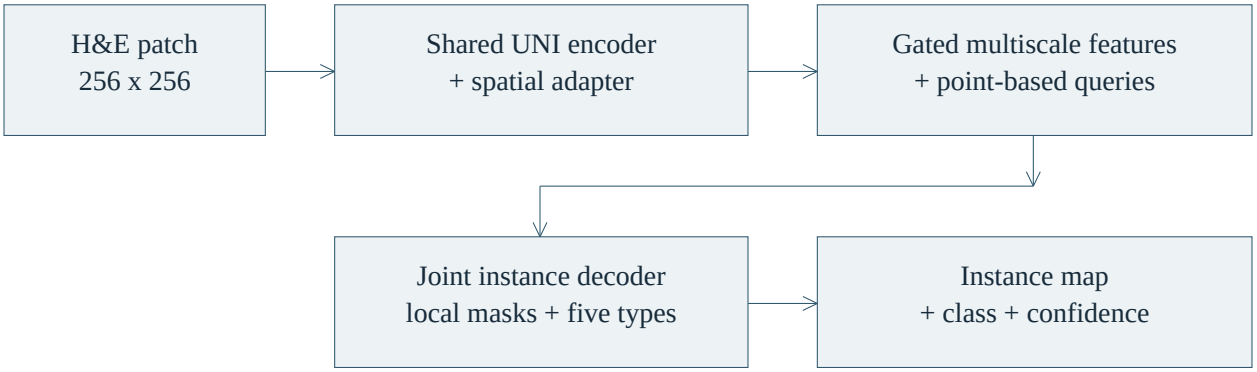
Mechanism analysis

The remaining analysis uses 2,523 fixed crop windows to measure misses among newly truncated Dead nuclei. A2-A1 isolates selective supervision, A2-A0 quantifies its overall tradeoff, and A3 tests whether the effect extends to all small instances. The frozen rule requires fewer misses in both seeds, a mean reduction of at least 0.03, and a positive 95% lower confidence bound. Additional Dead and strict-score conditions remain as specified in the original protocol [8].

Aim 3 tests whether direct mask prediction improves detection of partial or crowded nuclei. PM-Net replaces the HV decoder while retaining a matched encoder and training schedule, complementing Aim 2's controlled changes to supervision.

3. Aim 3: PM-Net, a direct instance model

PM-Net will predict a set of nucleus masks and cell types conditioned on candidate locations [9]. UNI provides tissue context, supplemented by a convolutional adapter for fine spatial features in small and crowded nuclei. The prototype will use native 256 x 256 patches and the same pretrained encoder initialization as CellViT-UNI [10].



Proposed PM-Net. One shared image encoding; no HV maps or watershed.

Architecture and optimization

Shared image features. A compact CNN adapter adds higher-resolution spatial features to UNI tissue representations, with learned gates combining the scales for the point and mask heads.

Point-conditioned queries. A supervised heatmap predicts candidate locations and types; each selected point initializes a feature vector, or query, for one proposed nucleus. Point targets lie inside visible instance masks so that cropped objects retain valid locations. Query capacity will be fixed from training-set object counts with a safety margin, initially about 256 per patch, and capacity overflow will be reported.

Joint mask and type prediction. Queries attend to local features and tissue context to predict masks, confidence, and five-class probabilities. One-to-one matching associates predictions with labeled nuclei by location, class, and mask agreement. Training combines point focal loss, mask cross-entropy and Dice losses, and type cross-entropy, with weights selected on validation data. Direct heatmap supervision provides a training signal for the point head despite non-differentiable point selection.

Instance decoding. A confidence rule selected on validation data resolves overlapping masks before mapping them to native image coordinates for scoring. We will assess how minimum-area filtering affects small-instance recall separately from the training-target cutoff in Aim 2.

Controlled ablations

A basic query decoder will be compared with versions that add point initialization and gated fusion separately. A plain adapter with matched parameter count will distinguish architectural effects from added capacity. The initial study retains five-class prediction throughout; truncation supervision and auxiliary type grouping are subsequent extensions rather than prerequisites for the core model.

4. Experimental design and evaluation

Development and confirmation

Development pairs seeds 19 and 1 on training fold 1 and validation fold 2, using the existing 130-epoch schedule and encoder unfreezing at epoch 25. The encoder initialization and training recipe will be matched across PM-Net and CellViT-UNI. Architecture, losses, augmentation, decoding, and decision criteria will be fixed before confirmation.

If development criteria and resources permit, confirmation will use train/validation/test folds 1/2/3, 2/1/3, and 3/2/1, evaluating final checkpoints for every split and seed before averaging. Splits 1 and 2 share test fold 3. Results will disclose prior exploratory use of test folds and the lack of identifiers needed to verify patient-level separation. Optional CoNIC and MoNuSAC evaluation will use class mapping and overlap exclusion.

Endpoints and statistical analysis

The shared PanNuke evaluator measures official mPQ, which includes type, bPQ, which ignores type, and per-class panoptic quality (PQ). Strict mPQ and strict Dead PQ additionally retain penalties for predictions of classes absent from an image. Detection and typing errors will be separated, with Dead recall stratified by tissue and by interior versus border location. Merge/split rates and tissue-by-class heatmaps will characterize crowded-instance performance.

Comparisons with matched CellViT-UNI and the crop-only control will use paired, tissue-stratified image bootstrap intervals. Crop windows from each source patch will form a single resampling cluster. Each seed will be reported separately, since these intervals measure image-sampling variation for the trained models. Class counts and annotation uncertainty will accompany rare-class estimates.

PM-Net development criteria

Before training, we propose fixing a primary target of at least 0.03 mean improvement in interior Dead recall, with a positive paired 95% lower confidence bound. Mean Dead PQ must improve by at least 0.005, and both recall and Dead PQ changes must be positive in each seed. Per-seed mPQ/bPQ losses are limited to 0.002; strict mPQ and strict Dead PQ must not decline, and mean merge/split rates must not increase. These criteria apply to PM-Net, independently of the frozen supervision experiment.

End-to-end inference, including decoding, is limited to twice the matched single-scale baseline latency. Repeated measurements on the same hardware and at the same batch settings will quantify latency, peak memory, and variability under shared-device load. A prototype meeting both quality and cost criteria will qualify for cross-split confirmation as resources permit.

5. Milestones, feasibility, and deliverables

Schedule	Milestone
Week 1	Complete the supervision analysis and baseline/error tables (minimum result).
Week 2	Finalize PM-Net design; implement and verify matching, masks, query capacity, and runtime.
Weeks 3-4	Complete the two-seed development comparison and core ablations (target result).
Weeks 5-6	Finalize the report; pursue cross-split and external evaluation if the prototype qualifies and resources permit.

The six-week schedule begins at proposal agreement, with three levels of scope:

Minimum result. Reproduced baselines, the completed four-arm supervision comparison, and reproducible error tables by tissue and cell class.

Target result. A tested PM-Net prototype and two-seed ablations that establish whether direct instance prediction improves Dead recall and PQ within the quality and latency criteria. A clear account of any remaining tradeoff is also a useful outcome.

Stretch goals. Cross-split confirmation of a qualifying prototype, external evaluation on CoNIC and MoNuSAC, and an exploratory auxiliary-supervision ablation. If the prototype falls short, Weeks 5-6 will focus on failure analysis and the final report.

Feasibility and main risks

Rare labels and annotation uncertainty. Tissue-specific counts and paired estimates will accompany rare-class results. Existing labels will be preserved, and disagreements among detectors will be retained as uncertain cases for future expert review.

Localization versus mask quality. In crowded regions, a correctly located query may still produce a mask with insufficient overlap for matching. Separating missed points, low-overlap masks, duplicates, merges, and splits will show whether localization or mask prediction limits performance.

Experimental scope and resources. The encoder and training recipe remain fixed during the decoder ablations. We estimate 2-4 A100 GPU-days for development and a further 4-8 for confirmation, subject to allocation. Available storage and checkpoint requirements will be checked before scheduling; resumable training will reduce losses from interruptions.

Deliverables and broader impact

The target deliverables are PM-Net code and trained models, versioned evaluation configurations, model/data provenance, comparison tables, and a final report with annotated failure cases. Together, these outputs will document how supervision and instance representation affect rare-nucleus detection under matched conditions.

This work supports more reliable measurements of cell composition across tissues. Reporting tissue-specific errors alongside aggregate scores will help researchers assess whether apparent improvements extend to underrepresented cells and justify their computational cost.

References and project evidence

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Versioned project documentation

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Frozen supervision specification

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Versioned new-model concept

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Team contributions to this proposal

This page is separate from the main proposal.

The team will complete the statements below before submission.

TODO - COMPLETE BEFORE COURSE SUBMISSION
Add each team member's actual contribution to this proposal.

This section is intentionally left for the team to complete. Record work on the proposal itself, such as developing the research question, reviewing literature, designing experiments, preparing figures, or drafting and revising sections.

Team member	Contribution to this proposal
Zhiyu Song	**TODO: add actual contribution.**
Yuchen Ai	**TODO: add actual contribution.**
Xinhao Jin	**TODO: add actual contribution.**
Zheng Jian	**TODO: add actual contribution.**